

CURVED 3-SEAT SOFA / CNC FRAME

15 mm PLYWOOD | mm | 1:1 DXF

1 - SPECIFICATION AND BILL OF MATERIALS

SPECIFICATION

Overall width (chord of the back)	2200 mm
Overall depth over the crescent	1074 mm
Radial depth of the seating band	900 mm
Plan sweep / back radius	60 deg / R2200 mm
Seat frame height (add cushion)	345 mm
Backrest height	760 mm
Arm height	620 mm
Material	15 mm birch plywood, 2440 x 1220 mm
Sheets required	6
Net part area	5.82 m2
Total cut path	114 m
Frame mass (600 kg/m3)	~52 kg
Joint clearance	slots 15.4 mm for 15 mm plate
Corner relief	dogbone R3.2 mm (6 mm cutter)

BILL OF MATERIALS

PART	QTY	SIZE (mm)	FUNCTION
RIB	7	900 x 760	radial rib, drops onto base rails
ARM-PANEL	2	900 x 760	end panel, slides on tangentially over the rail tabs
RAIL-BASE-IN	1	1413 x 273	z 0-15 mm, bottom_notch
RAIL-BASE-OUT	1	2173 x 375	z 0-15 mm, bottom_notch
RAIL-SEAT-IN	1	1403 x 272	z 315-330 mm, top_notch
RAIL-SEAT-OUT	1	1983 x 350	z 315-330 mm, top_notch
RAIL-BACK-BOT	1	2188 x 386	z 330-345 mm, outer_notch
RAIL-BACK-MID	1	2188 x 386	z 520-535 mm, outer_notch
RAIL-BACK-TOP	1	2183 x 376	z 745-760 mm, post_top_notch
SEAT-DECK-1	1	679 x 805	rests on ribs + both seat rails
SEAT-DECK-2	1	929 x 726	rests on ribs + both seat rails
SEAT-DECK-3	1	679 x 805	rests on ribs + both seat rails
BACK-STILE	14	70 x 430	threads down through the three back rails

TOTAL 33 parts

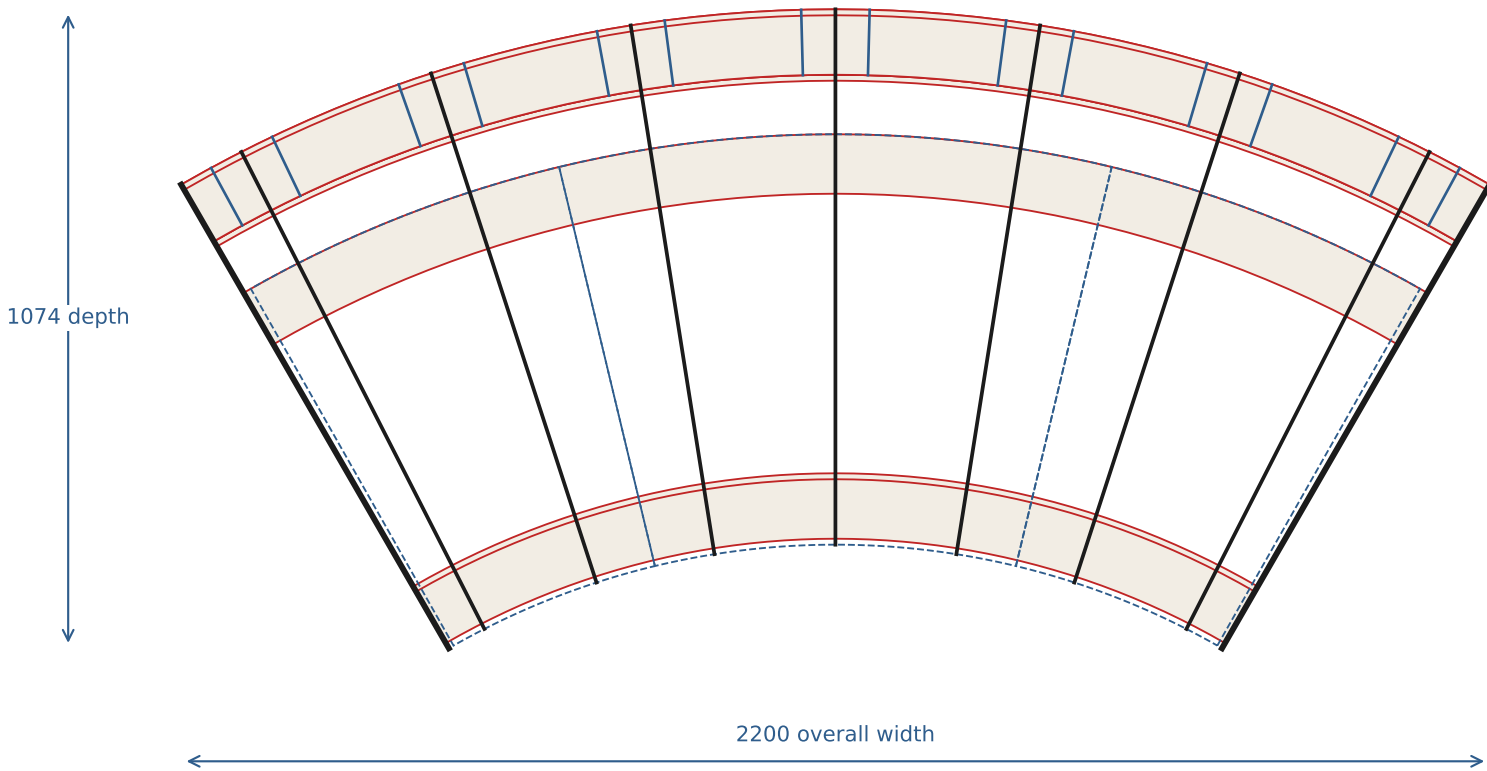
ASSEMBLY SEQUENCE

- Lay RAIL-BASE-IN and RAIL-BASE-OUT flat on the floor, concentric.
- Drop the 7 RIBs onto them; the notches in the rib feet fix every rib angle.
- Drop RAIL-SEAT-IN and RAIL-SEAT-OUT into the open notches in the rib tops.
- Drop RAIL-BACK-TOP into the notches in the tops of the back posts.
- Slide RAIL-BACK-BOT and RAIL-BACK-MID in from the outside of the posts.
- Slide the two ARM-PANELs on tangentially - the rail end tabs pass through them.
- Thread the 14 BACK-STILEs down through the three back rails; the 60 mm tenon seats in RAIL-BACK-BOT, the 70 mm shoulder lands on top of it.
- Drop the three SEAT-DECK sectors over the rib tabs; they bear on both seat rails.
- Glue every joint (PVA D3) and clamp. No fasteners are required.

CUTTING NOTES

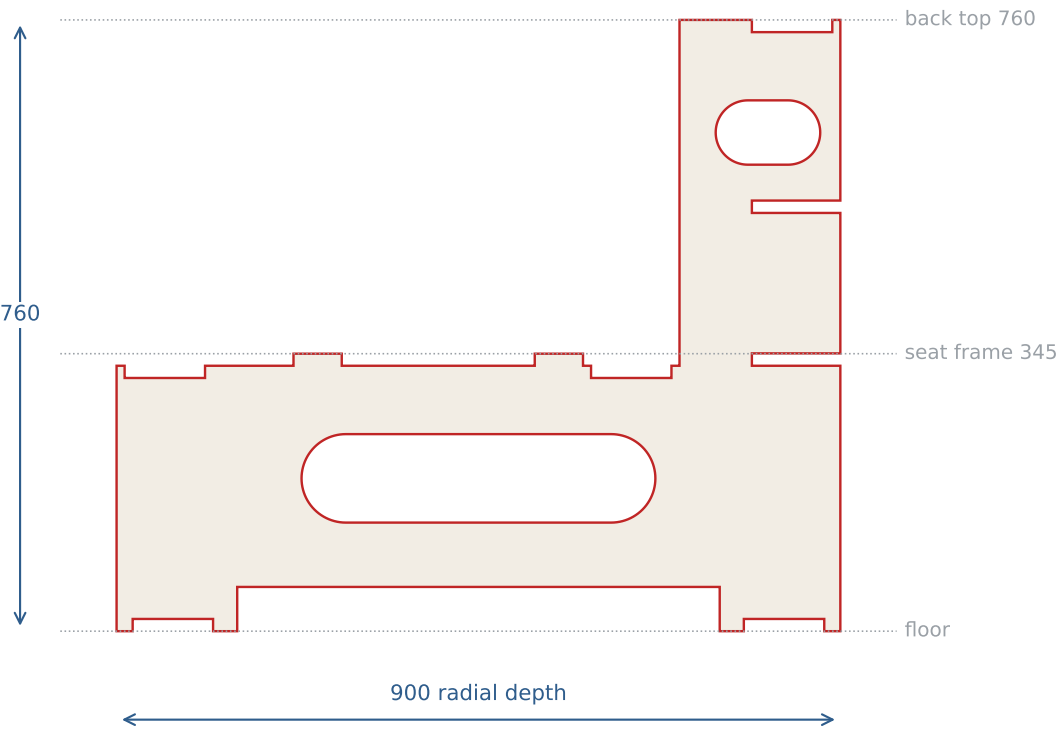
- Cut every CUT-layer contour through: internal contours first, outer profile last.
- 6 mm compression or up-cut spiral; climb finish pass; 0.1 mm onion skin or tabs.
- Dogbone relief is already in the geometry - do not add extra cutter compensation to the slots, only the usual tool-radius offset on the contour itself.
- Slots are cut 15.4 mm for 15.0 mm ply. Test one slot on an offcut before the run; if your sheet measures under 14.6 mm, edit FIT in sofa_geometry.py and regenerate.
- Grain direction: run the sheet grain along the length of the curved rails.
- Nesting is shelf-packed at ~33 percent sheet use; a true nesting optimiser can interleave the curved rails and typically saves one sheet.
- This is the frame only. Foam, webbing and fabric are not part of this cut file.

PLAN - sweep 60 deg, back R2200



solid lines: 7 ribs + 2 arm panels dashed: seat deck sectors

SECTION THROUGH A RIB (radial)



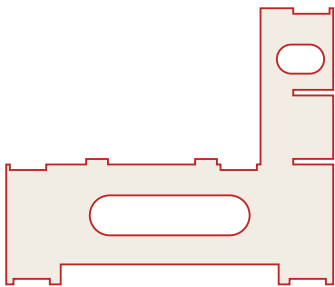
seat height with a 100 mm cushion: ~445 mm

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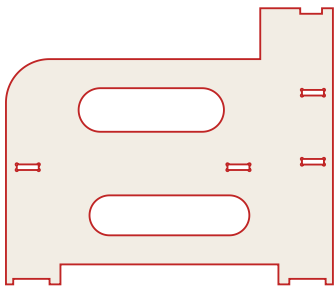
3 - PARTS INDEX - 13 distinct parts, 33 pieces

RIB x7



900 x 760 mm
sheet 3,4,5

ARM-PANEL x2



900 x 760 mm
sheet 5,6

RAIL-BASE-IN x1



1413 x 273 mm
sheet 2

RAIL-BASE-OUT x1



2173 x 375 mm
sheet 2

RAIL-SEAT-IN x1



1403 x 272 mm
sheet 3

RAIL-SEAT-OUT x1



1983 x 350 mm
sheet 2

RAIL-BACK-BOT x1



2188 x 386 mm
sheet 1

RAIL-BACK-MID x1



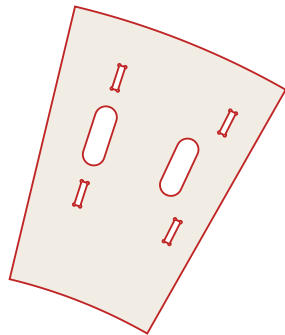
2188 x 386 mm
sheet 1

RAIL-BACK-TOP x1



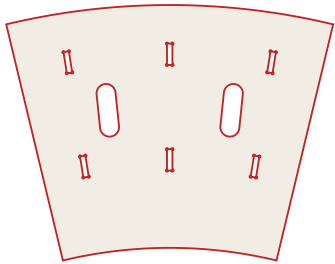
2183 x 376 mm
sheet 1

SEAT-DECK-1 x1



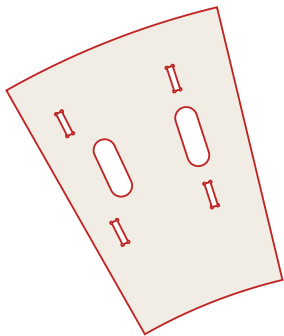
679 x 805 mm
sheet 6

SEAT-DECK-2 x1



929 x 726 mm
sheet 4

SEAT-DECK-3 x1



679 x 805 mm
sheet 6

BACK-STILE x14



70 x 430 mm
sheet 2

Each part is fitted to its own frame, so the drawings are NOT to a common scale - read the millimetre size under each one.
Every piece is 15 mm plywood; the sheet number tells you which nesting page it is cut from.
generated 2026-08-24 - curved-sofa.dxf.py - all dimensions in millimetres

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4 - NESTING SHEET 1 OF 6

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RAIL-BACK-BOT x1, RAIL-BACK-MID x1, RAIL-BACK-TOP x1

